BAMNET 2025: interpreting the detection output

1-minute data: 187 cold-pool candidates; 148 temperature-only; 4,117 wind-only.

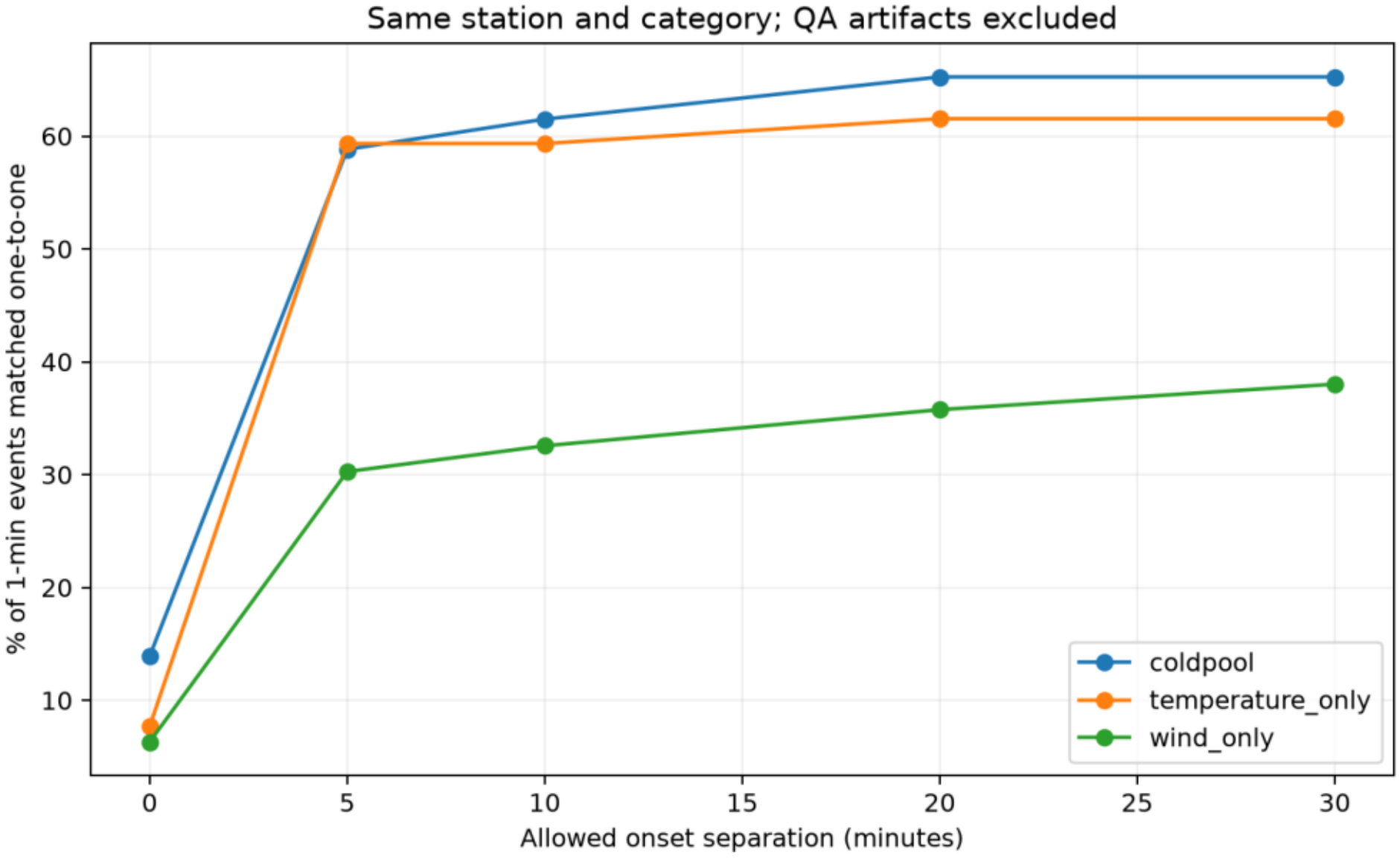
The 148 temperature-only entries contain 91 valid thermal episodes
and 57 end-of-file missing-mask artifacts retained by legacy mode.

Wind-only entries are not additional cold pools. Temperature-only
does not mean that the wind remained unchanged.

5-minute data: 197 cold-pool candidates; 178 temperature-only; 2,294 wind-only.
The 178 temperature-only entries contain 121 valid episodes and 57 QA artifacts.

Physical interpretation excludes QA artifacts; original catalogs remain available.
These are station-derived candidates without satellite verification.

Event overlap: compare matched episodes, not just totals



Same station, one-to-one maximum-cardinality matching with minimum total onset separation; QA excluded.
At ±20 minutes: 122 cold-pool matches, representing 65.2% of the 1-minute and 61.9% of the 5-minute candidates.

What does monotonic decrease mean in the code?

```
CODE: monotonic = (window.diff("window") <= 0).all("window")

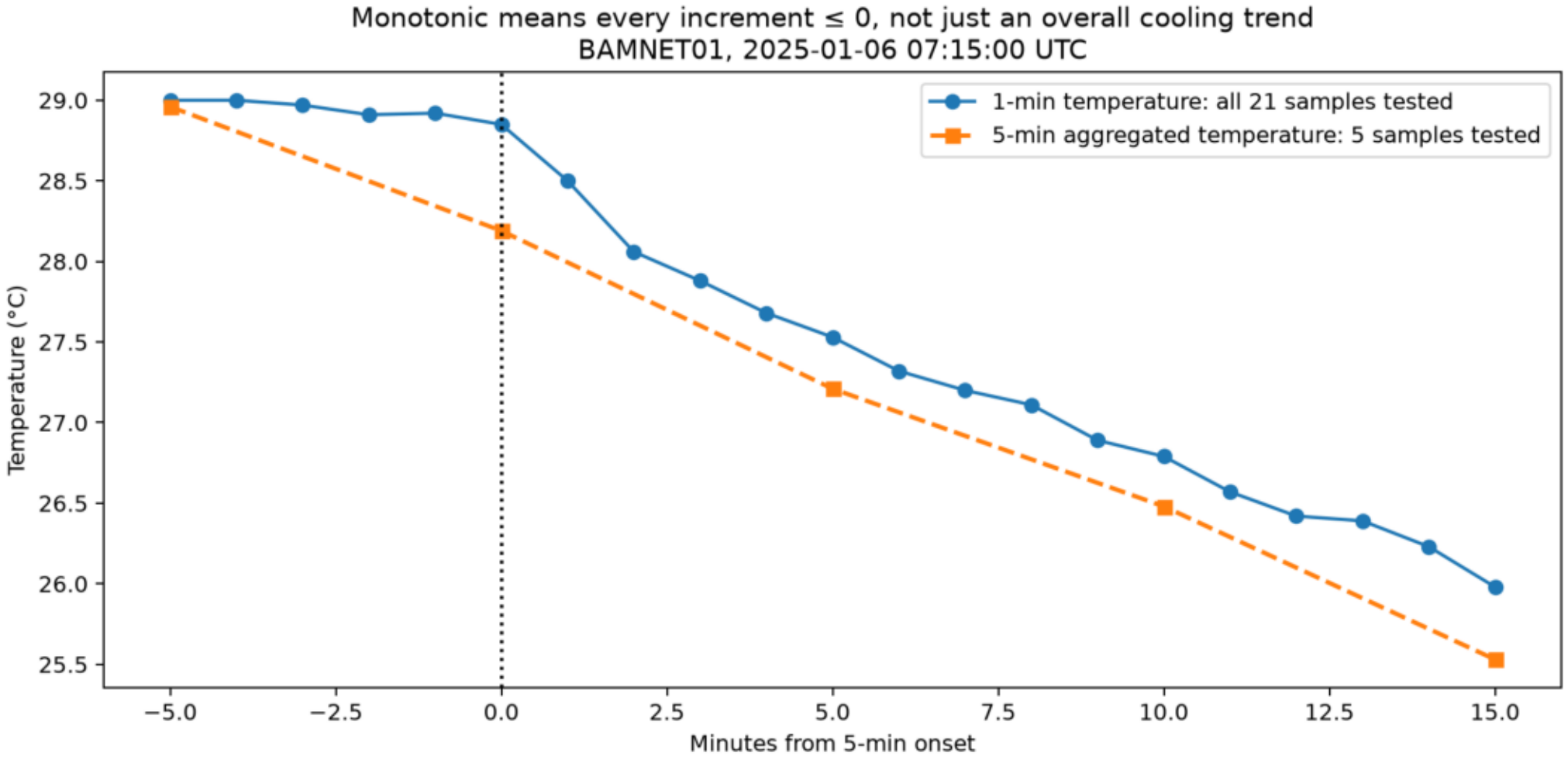
Meaning: every consecutive temperature increment must be <= 0.
Equal values are allowed; a small warming step fails.
This is stricter than an overall downward trend or a negative slope.

5-min data: 5 temperatures, 4 increments over 20 minutes.
1-min data: 21 temperatures, 20 increments over 20 minutes.

Additional conditions: total drop >=2 C; first 5-min drop >=0.5 C.
```

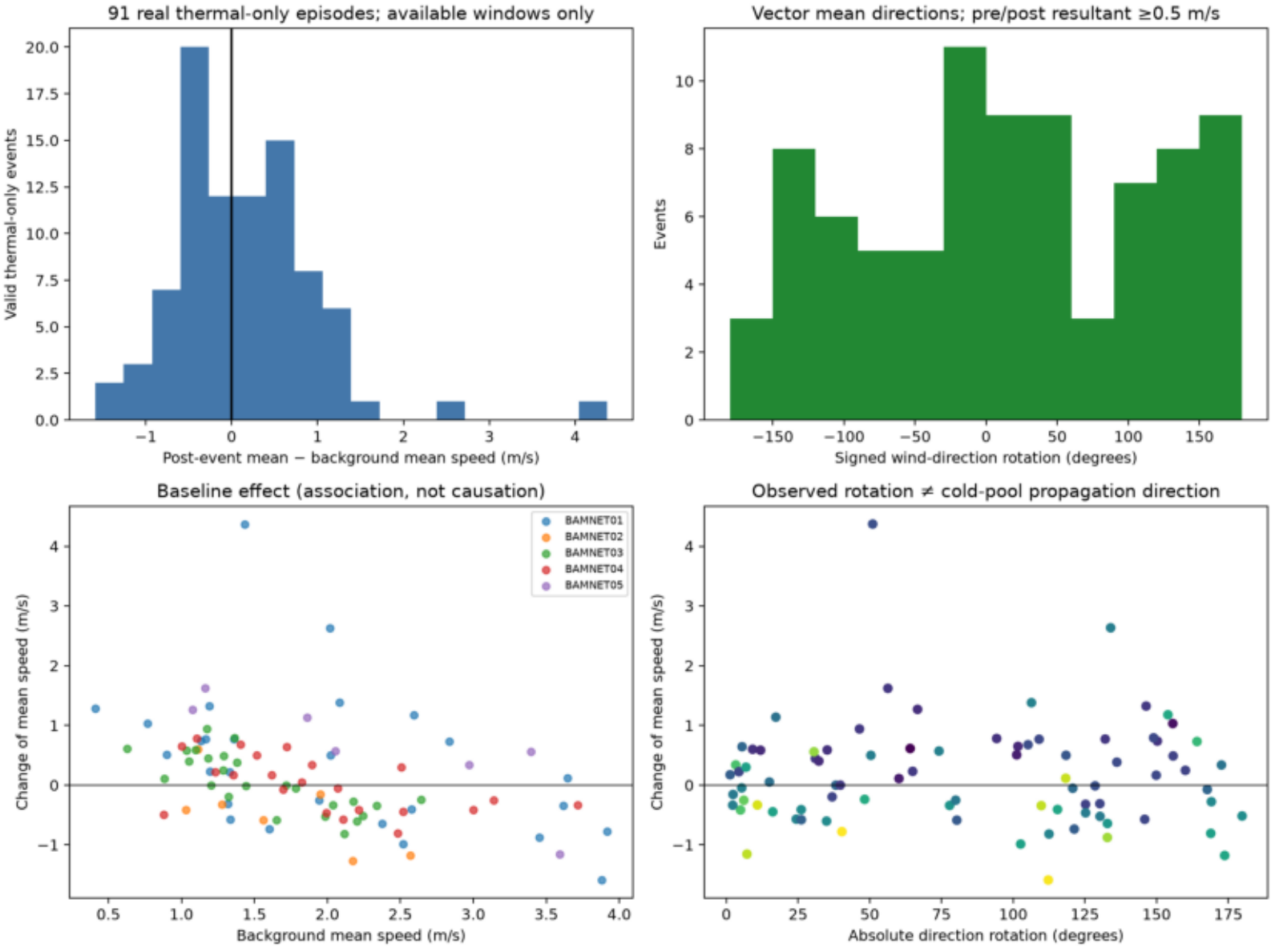
The actual implementation checks every consecutive increment within the rolling window.
Equal temperatures are allowed; any positive increment fails. This is stricter than a generally downward trend.

A real example: aggregation changes the monotonicity test



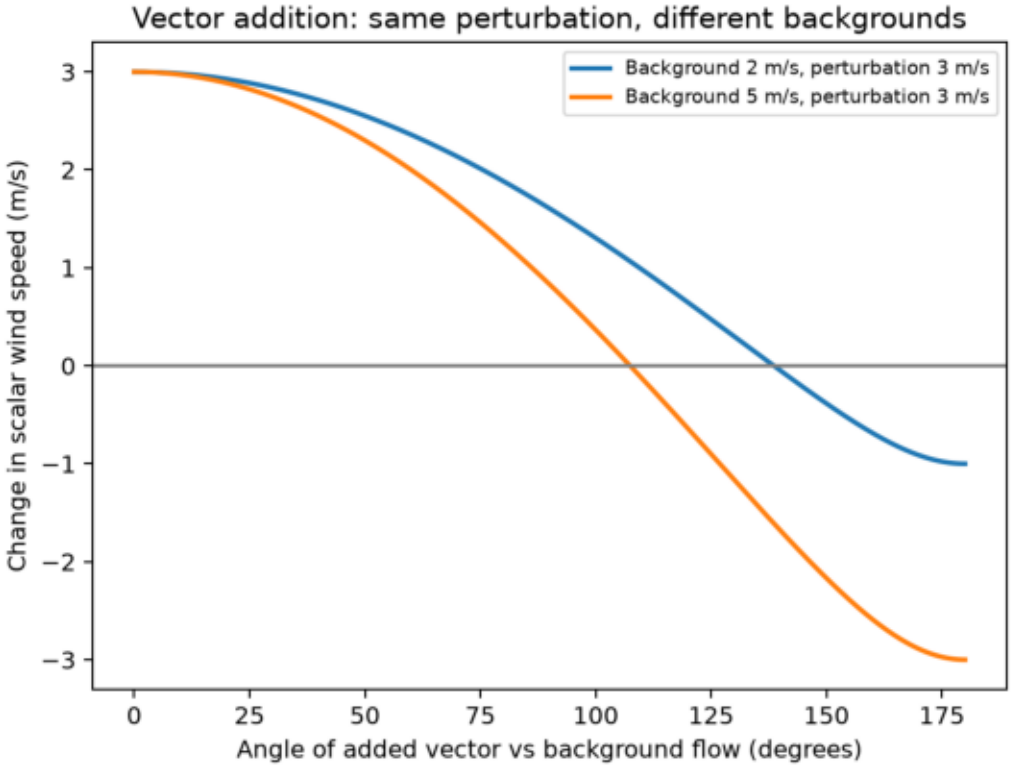
BAMNET01, 6 January 2025: a +0.01°C minute-scale increase fails the test at this particular anchor.
Another 1-minute anchor may still qualify. The 5-minute series is an aggregation, not simple subsampling.

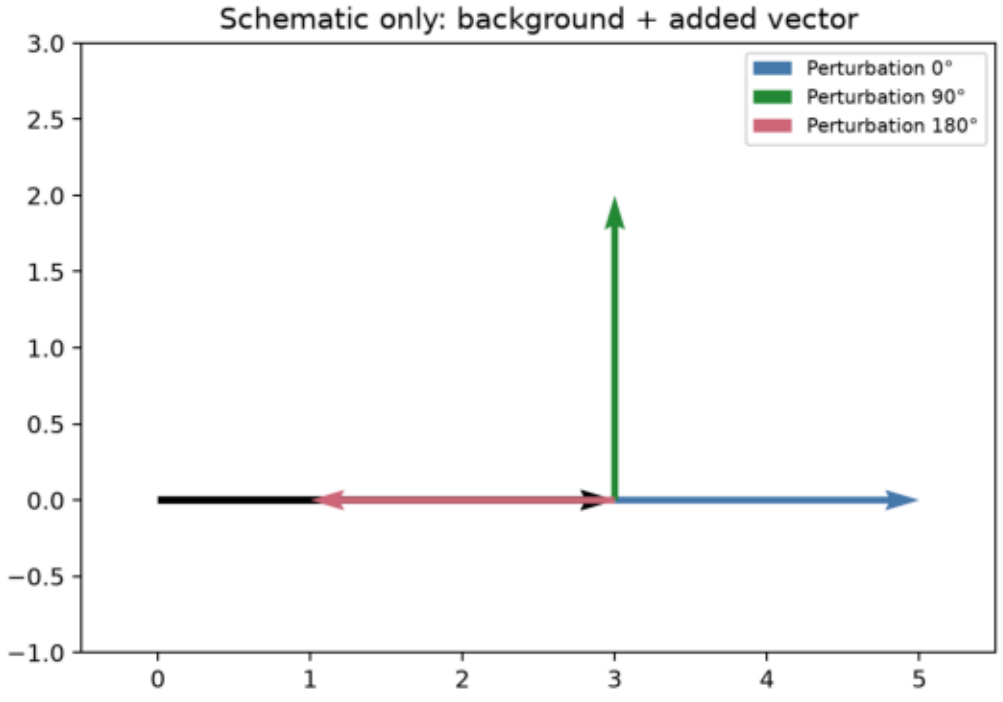
Temperature-only episodes: speed, direction, and background



Among 91 valid episodes: mean speed increased in 45, decreased in 42, was unchanged in 1; 3 lacked sufficient data.
Background: −40 to −20 minutes. Event window: 0 to +20 minutes. Each mean requires at least 17 of 21 samples.

Vector geometry: aligned, perpendicular, and opposing flow





This is a theoretical illustration, not a measured propagation direction. An opposing perturbation can reduce speed or reverse the flow. Local wind direction and cold-pool propagation direction are different quantities.

Background speed and direction: descriptive evidence

88 valid thermal-only episodes have sufficient pre/post speed data.
Median mean-speed change: +0.081 m/s; interquartile range: −0.415 to +0.591 m/s.

83 episodes have reliable pre/post vector-mean directions (both magnitudes ≥0.5 m/s).
Median absolute rotation: 80.4°. Rotations: 29 below 45°, 37 from 45–135°, 17 above 135°.

Background speed versus mean-speed change: Spearman rho = −0.510; $p = 3.97 \times 10^{-7}$.
The fraction with increasing speed falls from 86.4% in the lowest background quartile
to 27.3% in the highest quartile.

This is not causal evidence: the change already subtracts background speed,
creating mathematical coupling and possible regression to the mean.

The relationship with peak-speed increase is weaker: rho = −0.202; p = 0.059.
P-values are descriptive and assume event independence.

How the composite is normalized: onset subtraction

For each event i and variable X, the right-hand panels use:

$$X'_i(t) = X_i(t) - X_i(0)$$

Subtract the value at onset separately for each event, then compute the ensemble mean and standard deviation at each relative minute.

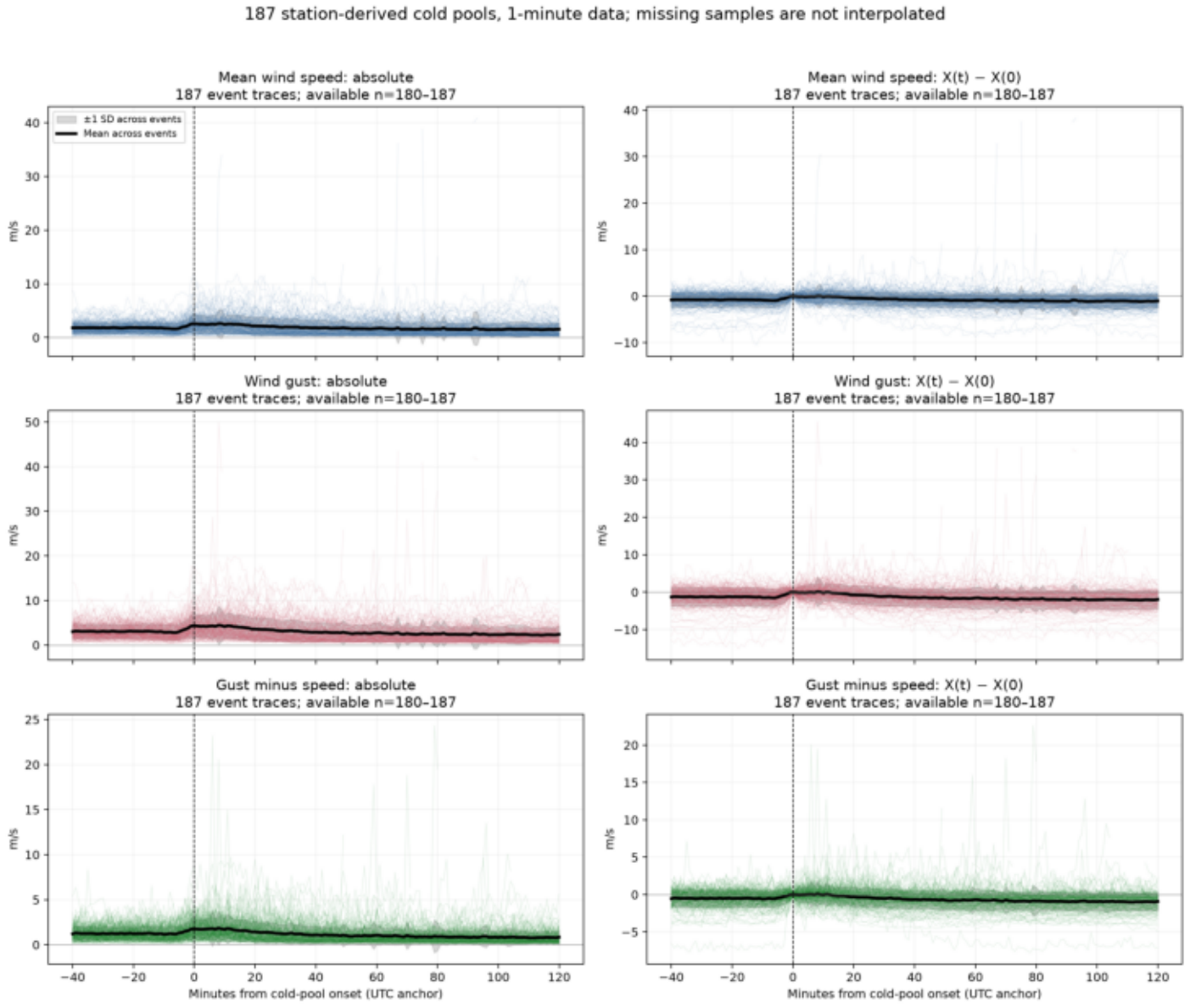
Example: event A changes from 5 to 7 m/s; event B changes from 2 to 4 m/s. Both become 0 m/s at onset and +2 m/s after the increase.

For the difference panel, D(t) = gust(t) − speed(t), then D′(t) = D(t) − D(0). The left-hand panels retain the original absolute values.

Units remain m/s. This is neither division, percentage change, nor a z-score. Variability is not standardized, and all onset-centered traces equal zero at t=0.

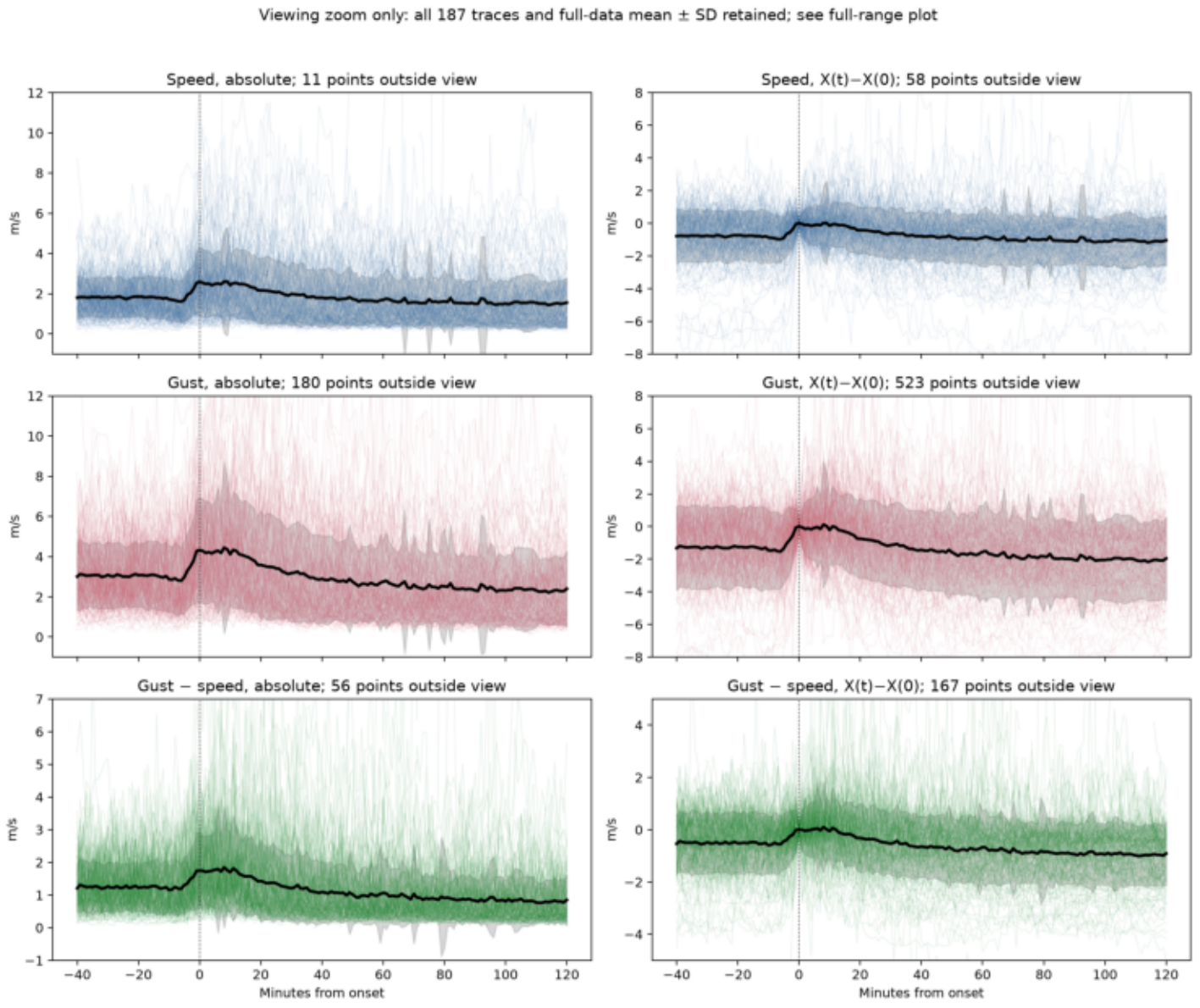
A gust can precede thermal onset, so the onset value may already be elevated. A flat or decreasing post-onset curve does not imply the absence of a gust front.

Requested six-panel composite: all 187 cold-pool candidates



Rows: 1-minute mean speed, 1-minute maximum gust, gust minus speed. Left: absolute; right: X(t)−X(0).
Each panel: 187 event traces, thick black mean, and ±1 population SD across events (not a confidence interval).
Window: −40 to +120 minutes; no interpolation. Available samples per minute: 180–187.

Viewing zoom: retain the data and disclose the clipping



Only the axis view is restricted; all data remain in the means and standard deviations.
Eight extreme-sample review flags occur in five events. Flags do not automatically invalidate or remove observations.

Wind speed or gust: a provisional decision

186 cold-pool candidates have sufficient paired pre/post speed and gust data.

Median peak increase above background: mean speed +1.872 m/s; gust +3.135 m/s.
Median change in event-window mean: speed +0.387 m/s; gust +0.701 m/s.

The fraction with an increasing event-window mean is similar:
67.7% for speed versus 68.3% for gust.

Gust has a larger response amplitude, but gust is an interval maximum
and this sample was selected using a gust criterion. Selection bias is present.

Provisional choice: retain gust for consistency with the detection method;
use mean speed and u/v components to study the wind response and direction.

Higher detection accuracy is not established. Compare both indicators on the
same thermal candidates, with independent labels, non-events, and spatial/satellite checks.